

A Percent Needs Its Base

AP Statistics · Unit 1 · Topic 1.3 · TEACHER KEY

CED TETHER

- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 2.A** — Describe data presented numerically or graphically.
- **Enduring Understanding UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.A** — Represent categorical data using frequency or relative frequency tables. [Skill 2.B]
- **EK UNC-1.A.1** — A frequency table gives the number of cases falling into each category. A relative frequency table gives the proportion of cases falling into each category.
- **LO UNC-1.B** — Describe categorical data represented in frequency or relative tables. [Skill 2.A]
- **EK UNC-1.B.1** — Percentages, relative frequencies, and rates all provide the same information as proportions.
- **EK UNC-1.B.2** — Counts and relative frequencies of categorical data reveal information that can be used to justify claims about the data in context.

Essential question: When does a count answer the question, and when does a percent — and what does each one hide?

DOK-3 skill: 4.B · **FRQ pattern:** frequency-vs-relative-frequency-which-table-the-argument-needs

DOK rationale: Part (c) chooses a meaning for interest, matches counts or percentages to that meaning, and explains why unequal club totals can reverse the comparison.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- If the session room holds 30 chairs, which table tells the council how many chairs each club fills?
- 83% of what? 60% of what? Say the base out loud.
- Why did Robotics choose counts and Chess choose percents? What would each look like on the other table?
- Rewrite the council's question so that the *other* table is the right one to answer it.

[WARN] LOOK FOR

- 'Percents are always better' (or 'counts are the real data') with no link to what the council asked
- Cites 83% without ever saying it is 15 people out of 18
- Compares 27 Yes to 15 Yes without noting Robotics has 45 members to Chess's 18

SCORING PART (c) — E / P / I

Expected elements

- **defines-interest** (required) — Chooses a club with a stated people-versus-share interpretation

- **uses-matching-numbers** (required) — Uses 27 versus 15 or 83 percent versus 60 percent consistently
- **explains-reversal** (required) — Explains that different club totals reverse the lead and should accompany the percents

E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

Common mistakes

- ‘Percents are always better’ (or ‘counts are the real data’) with no link to what the council asked
- Cites 83% without ever saying it is 15 people out of 18
- Compares 27 Yes to 15 Yes without noting Robotics has 45 members to Chess’s 18
- Reports 27/45 as ‘27%’ — treats a count as a percent
- Rounds 15/18 to 80% or 85%

[STAR] THE PROBLEM — annotated

Two clubs asked the student council for money to run a Saturday session. Each club surveyed *all* of its members: “Would you attend a Saturday session?” The table shows every response as a count.

At the council meeting, the **Robotics club** presented a **frequency table** (counts) and the **Chess club** presented a **relative-frequency table** (percents). Each club chose the table that made its case look best.

“Would you attend a Saturday session?” (every member surveyed; counts)

Club	Yes	No	Total
Chess	15	3	18
Robotics	27	18	45

FIRST TAKE — one sentence before you work the parts

Which club has more interest in a Saturday session — Chess or Robotics? One sentence and one number.

Key: Not graded. Expect a split: some say Robotics (27) and some say Chess (83%); the split IS the lesson.

[BOOK] TABLES FOR A CATEGORICAL VARIABLE (keep on the page)

A **frequency table** gives the count in each category. A **relative-frequency table** gives its proportion or percent: $\text{relative frequency} = \frac{\text{category count}}{\text{group total}}$. Multiply by 100 for a percent. Counts answer ‘how many?’, percents answer ‘what share?’. Always identify the group that a percent describes.

DOK 1 (a) How many Robotics members said Yes?

Key: 27 Robotics members said Yes.

DOK 2 (b) Find the percent saying Yes in each club (nearest whole percent). Which is higher?

Key: Chess: $15/18 \approx 83\%$. Robotics: $27/45 = 60\%$. Chess has the higher percentage saying Yes.

DOK 3 (c) Which club has more student interest? State whether you mean more people or a larger share, and use the matching numbers. Explain why the other measure favors the other club and what a reader needs to know to compare them fairly.

Key: If interest means number of people, Robotics leads 27 to 15; if it means share of members, Chess leads about 83% to 60%. Either interpretation earns credit when stated. The lead switches because Robotics has 45 members and Chess 18. Report the club totals with the percentages so readers can distinguish the number of interested students from the share. The full count table includes the totals; it does not hide them.